PART 2

Airway and Breathing Emergencies

IN THIS PART

1. Identify the structures of the upper airway and what narrows or blocks them.
2. Recognize partial and complete airway obstruction and state the response.
3. Describe basic airway positioning, suction and bag-valve-mask ventilation, and their safety limits.
4. Recognize respiratory distress and respiratory failure and state first priorities for asthma, chronic obstructive pulmonary disease (COPD) and pneumonia.
5. Recognize anaphylaxis and state the first priorities.

A patient can go for days without food, but only minutes without air reaching the lungs and the brain. That is why airway and breathing come first in the primary survey of Entry 1.3, and why this Part comes next.

The five entries follow the path that air takes and the ways that path fails. Entry 2.1 describes the passage itself and how to judge whether it is open. Entry 2.2 covers what to do when it is not, from simple positioning to clearing a choking adult. Entry 2.3 explains how to move air for a patient who cannot, with a bag-valve-mask and supplemental oxygen. Entry 2.4 turns to the lungs and the

conditions that make breathing hard. Entry 2.5 ends with anaphylaxis, which can close an airway within minutes and which a bystander may be the first to treat.

Everything here assumes the habits of Part 1: a safe scene, a first look, and a recheck after every action. As before, what you may do depends on your scope of practice and local protocol. Where a skill needs training that a book cannot give, the text says so plainly and tells you when to stop and call for help.

2.1 Airway Anatomy and Assessment

The airway is the passage that carries air from the outside to the lungs. You cannot judge whether it is open, or why it has closed, without a picture of what it is made of. This entry gives you the picture and then the short set of observations that tell you how well it is working.

WHAT IT IS. Air enters through the nose and mouth and passes into the throat, called the pharynx. Below the throat sits the larynx, or voice box, guarded by a flap of tissue called the epiglottis that covers the entrance when you swallow. Below the larynx the windpipe, or trachea, runs down into the chest and divides into the two main bronchi, which branch into ever smaller tubes inside the lungs. The part from the nose down to and including the larynx is the upper airway, and most emergency airway care is aimed there, because it is the part you can reach and open.

Figure 5 shows these structures; notice the order from nose to windpipe.

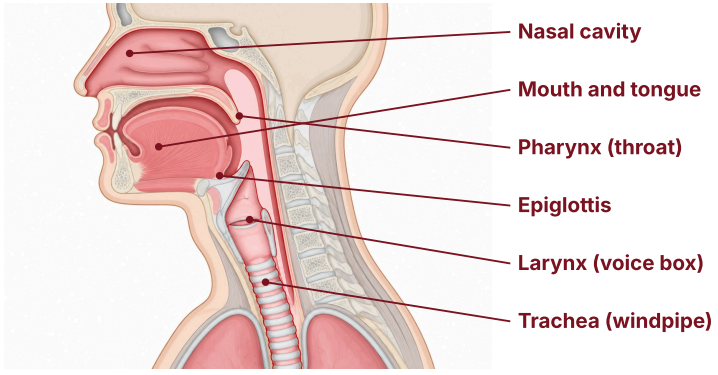


Figure 5. The upper airway from nose and mouth to the windpipe, with the main structures labelled.

WHY IT MATTERS. The upper airway is narrow, shared with the route for food and drink, and held open by muscle. Anything that weakens the muscle, adds swelling, or puts something in the way can close it. Because the brain depends on a constant supply of oxygen, a blocked airway can cause loss of consciousness and then cardiac arrest if it is not cleared.

WHAT NARROWS OR BLOCKS IT. Five causes are common. In an unconscious patient lying on the back, the tongue and soft tissues of the throat can fall backward and close the passage, and this is the most common cause of obstruction in an unconscious patient. Food or a foreign object can lodge in the throat. Fluid, vomit or blood can fill the airway. Swelling from an allergic reaction, infection, a burn or an injury can narrow it from within. A direct injury to the face or neck can distort it. Each cause points to a different first action, which is why the next entry is organized by cause and not by technique.

ASSESSING THE AIRWAY. Entry 1.3 introduced looking, listening and feeling; here each finding is tied to a cause. Ask the patient a simple question. A patient who answers in a normal voice has an open airway for the moment. Then watch the chest and abdomen, listen for sounds, and feel for air at the mouth and nose.

What airway findings suggest

Finding	What it suggests
Normal voice, full sentences	Airway open for now
Very hoarse or whispered voice	Swelling or narrowing of the throat
High-pitched sound on breathing in (stridor)	Significant partial blockage
Gurgling	Fluid in the airway
Chest and abdomen moving in opposite directions	Obstruction with strong effort to breathe
No air felt or heard at the mouth or nose	Complete obstruction or no breathing

Airway obstruction is an emergency. In a partial obstruction, air still moves but less than it should and often with noise. In a complete obstruction, no air moves, the patient cannot speak or cough, and the chest and abdomen may move in opposite directions as the patient strains. A blue or gray tint to the lips, called central cyanosis, is a late sign. Do not wait for it.

RED FLAGS

A very hoarse or whispered voice, a high-pitched sound on breathing in, drooling or inability to swallow, swelling of the face or tongue, a patient who cannot speak or cough, or a blue or gray tint to the lips. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Assess the airway first and again after every change in the patient, and treat obstruction as an emergency without waiting for a blue tint.

See also: Entry 1.3 (primary survey); Entry 2.2 (opening and clearing the airway); Entry 6.1 (child airway differences).

2.2 Airway Obstruction and Basic Airway Management

The simple measures in this entry can be done with few or no tools, and they address the most common airway problems before advanced help arrives. They follow the cause. If the tongue has fallen back, you reposition. If there is fluid, you suction. If there is a solid object, you relieve choking. If there is swelling, you call for help and treat the cause, as Entry 2.5 describes for allergic swelling.

WHAT IT IS. Basic airway management is the set of non-invasive actions that open and protect the airway without placing any device inside the patient. It includes positioning the head and jaw, clearing the mouth with suction, placing a patient who is breathing but unresponsive on their side, and relieving choking.

WHY IT MATTERS. Each measure is quick and each can be done by a student with the right training and a supervisor present. They matter because they act on the cause directly and because

they must be repeated: an airway opened once may close again as the patient's condition changes.

OPENING THE AIRWAY. The two standard positions lift the tongue away from the back of the throat. In the head-tilt chin-lift, you tilt the forehead back with one hand and lift the chin with the fingertips of the other. It is used for a patient with no suspected neck injury. In the jaw thrust, you place your fingers behind the angles of the jaw on both sides and lift the jaw forward without tilting the head. It is the choice when a neck injury is possible, because it moves the neck less. If the jaw thrust does not open the airway, your protocol may direct you to use the head tilt anyway, since an open airway comes first. Entry 4.3 returns to the neck.

Figure 6 compares the two positions; notice that the jaw thrust leaves the head in line with the body.

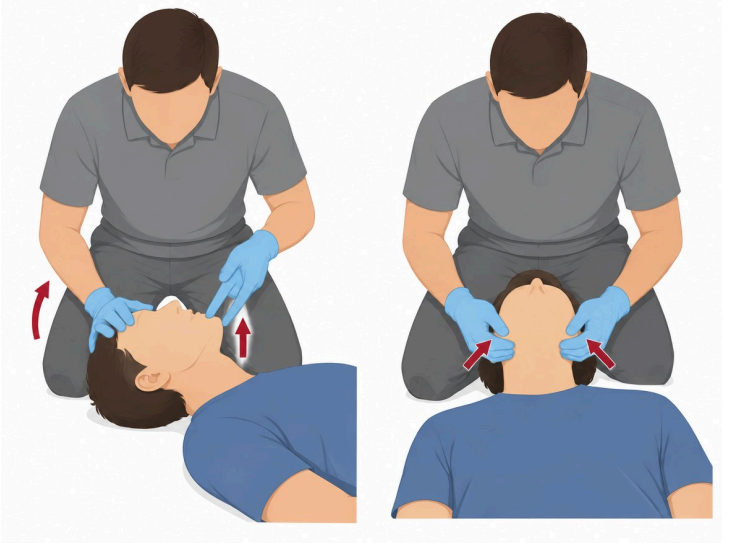


Figure 6. The head-tilt chin-lift and the jaw thrust, drawn side by side to show where the hands go.

CLEARING THE MOUTH. Suction removes liquid and small particles from the mouth and throat. Use it only when you can see or hear fluid, apply it in short bursts, and stop to check the patient's breathing and condition between bursts, because suction can lower the patient's oxygen level. Follow the device maker's instructions and your protocol for how long and how deep.

PROCEDURE: OPEN THE AIRWAY OF AN UNRESPONSIVE PATIENT

Purpose: to keep air moving past the tongue. Scope: follows your training level and local protocol.

1. Make the scene safe, put on gloves, and call for help.
2. Check for a response and for normal breathing (Entry 1.3).
3. If you suspect a neck injury, use the jaw thrust; otherwise, tilt the head and lift the chin.
4. Look in the mouth and clear anything you can see and reach, such as fluid or loose debris, with suction if it is available and you are trained.
5. Look, listen and feel for breathing again after the airway is open.
6. If the patient is breathing normally and you do not suspect an injury, place them on their side in the recovery position; if you suspect an injury, leave them in the position you found them, hold the airway open with the jaw thrust, and move them only as your protocol directs.
7. If the patient is not breathing normally, begin CPR (Entry 3.4).
8. Reassess and keep checking breathing until help takes over.

> **⚠ SAFETY**
>
> Do not put your fingers blindly into a patient's mouth. Stay with an unresponsive patient, and do not move a patient with a possible neck injury unless the scene is unsafe or your protocol directs a move to give urgent care, and then keep the head, neck and body in line as far as you can.

STOP AND CALL FOR HELP IF the airway does not open, if suction does not clear the mouth, or if the patient stops breathing.

These steps work in a fixed order because each one answers what the last one found. Positioning solves the common problem first, the fallen tongue. Clearing the mouth then deals with fluid or debris. Checking breathing then tells you whether the patient needs only an open airway or needs breathing support as well.

THE RECOVERY POSITION. A patient who is breathing normally but is unresponsive, and who has no suspected neck or spinal injury, is at risk from their own tongue and from vomit. Placing them on their side lets fluid drain from the mouth and keeps the tongue forward. Kneel beside the patient, place the arm nearest you at a right angle to the body, bring the far arm across the chest and hold the back of that hand against the near cheek, bend the far knee, and roll the patient toward you by pulling on that knee. Adjust the top leg so that the hip and knee are bent at right angles and tilt the head back slightly so the airway stays open. Check breathing again at once and keep checking. The position is for a patient who is breathing normally; a patient who is not breathing normally needs CPR, and a patient with a possible spinal injury is moved only as your protocol directs.

CHOKING IN AN ADULT. Choking is a solid object blocking the airway. A patient with a partial blockage can still cough, speak or breathe with effort, and you should encourage coughing and stay close and do not interfere while the cough is forceful. A patient with a severe blockage cannot speak, cough or breathe, may clutch the throat, and may turn blue or gray. That patient needs immediate help. Ask "Are you choking?" and, if they nod and cannot speak, act.

PROCEDURE: RELIEVE CHOKING IN A CONSCIOUS ADULT

Purpose: to dislodge an object that blocks the airway. Scope: follows your training level and local protocol; the exact sequence your service teaches takes precedence.

1. Ask "Are you choking?" and check whether the person can cough, speak or breathe.
2. Send someone to call emergency services, or call for help yourself as soon as you can.
3. Stand to one side and slightly behind the person, support the chest with one hand, and lean them forward.
4. Give up to 5 firm blows between the shoulder blades with the heel of your other hand.
5. If the object is still stuck, stand behind the person, place a fist just above the navel, and give up to 5 quick inward and upward abdominal thrusts.
6. Keep alternating back blows and abdominal thrusts until the object comes out or the person stops responding.
7. If the person becomes unresponsive, lower them carefully to the ground, call for help, and begin CPR, starting with chest compressions (Entry 3.4).

8. Stay with the person, keep watching them until help arrives, and have a clinician check them, even if the object comes out.

> **⚠ SAFETY**
>
> Abdominal thrusts can injure internal organs. For a person who is in the late stages of pregnancy or too large to reach around, your training may direct chest thrusts instead. Do not use abdominal thrusts on infants; Entry 6.2 covers them.

STOP AND CALL FOR HELP IF the object does not come out, the person cannot breathe or turns blue or gray, or the person becomes unresponsive.

The two techniques use different forces. Back blows strike the upper back to dislodge the object, and abdominal thrusts raise the pressure in the chest so that air pushes the object upward. Current guidance has you alternate them until the object comes out or the person becomes unresponsive.

NURSING AND RESPONDER PRIORITIES. Match the action to the cause, open the airway first, and suction only when you can see or hear fluid. Stay with an unresponsive patient and recheck the airway after every move.

See also: Entry 1.3 (primary survey); Entry 3.4 (CPR); Entry 4.3 (spinal motion restriction); Entry 6.2 (infant and child choking).

2.3 Ventilation Support: Bag-Valve-Mask and Oxygen Delivery

Sometimes the airway is open and the patient still is not moving enough air. Breathing may be too slow, too shallow or absent. At that point a trained responder moves air for the patient with a bag-valve-mask, often connected to oxygen. This entry explains

the device, the steps, and the oxygen devices you may meet. It names the device types and when each is used; your protocol sets the flow.

WHAT IT IS. A bag-valve-mask, written BVM, has three parts: a soft face mask that seals over the nose and mouth, a one-way valve, and a self-refilling bag that you squeeze to push air into the lungs. Released, the bag refills from room air or from an attached oxygen supply. It is used on patients who are not breathing or are breathing too weakly, and it works best with two trained people, one to hold the mask and one to squeeze the bag.

WHY IT MATTERS. Moving air by hand is a skill that takes practice. A poor seal lets air escape. Squeezing too hard or too fast can push air into the stomach which raises the risk of vomiting. A technique that moves the chest well on a manikin may fail on a patient with a beard or a swollen face. Practice on manikins, under supervision, comes first.

USING A BAG-VALVE-MASK. Choose a mask that covers the nose and mouth without pressing on the eyes. With two people, one holds the mask with both hands, keeping the airway open with the jaw thrust or head tilt, while the other squeezes the bag. With one person, the single hand and the bag are harder to manage, and the seal is the usual point of failure. Squeeze gently and just enough to make the chest rise, then release and let the chest fall. Watching the chest tells you whether air is going in.

Figure 7 shows the two-hand mask seal; notice how the fingers lift the jaw toward the mask.



Figure 7. A two-hand mask seal with a second person squeezing the bag (eye protection not drawn).

For an adult who has a pulse but is not breathing normally, a common rule is 1 breath about every 6 seconds, or about 10 breaths a minute. Your protocol sets the rate you follow, and the rate for a child or infant differs.

PROCEDURE: ASSIST BREATHING WITH A BAG-VALVE-MASK

Purpose: to move air into the lungs of a patient who is not breathing adequately. Scope: follows your training level and local protocol; usually done under supervision.

1. Make the scene safe, put on gloves and eye protection, and call for help.
2. Open the airway and clear the mouth (Entry 2.2).

3. Check for breathing and a pulse; if there is no pulse, begin CPR (Entry 3.4).
4. Choose a mask that covers the nose and mouth, and attach the bag and, if available, the oxygen supply.
5. Hold the mask on the face with both hands, with the jaw lifted, so that no air leaks.
6. Have the second person squeeze the bag gently until the chest begins to rise, then release.
7. Give breaths at the rate your protocol sets, watching the chest rise and fall.
8. Keep the airway open, reassess often, and stay with the patient until help takes over.

> ⚠ **SAFETY**
>
> Do not over-squeeze or give breaths too fast. If the stomach swells or the patient vomits, turn the patient on the side if you can, clear the mouth, and call for help. Never use oxygen near flame, sparks or smoking.

STOP AND CALL FOR HELP IF the chest does not rise, the patient vomits, or you cannot make a seal.

IS THE VENTILATION WORKING? The chest should rise gently and fall fully as the bag is released. You should see no air escaping around the mask. A good response may include improved skin color and, in a patient with a pulse oximeter, a rising reading, though that takes time. If the chest does not rise, reposition the head and jaw, reseat the mask, and check for an obstruction before squeezing harder. Squeezing harder risks inflating the stomach.

OXYGEN DELIVERY DEVICES. Oxygen can be given by several devices, chosen by how much the patient needs and by the patient's ability to breathe. A nasal cannula is a soft tube with two short prongs that rest in the nostrils, used for patients who are breathing well and need a small boost. A simple face mask covers the nose and mouth and is used when a moderate increase is needed. A non-rebreather mask has a bag, called a reservoir, that fills with oxygen so that the patient breathes in a high concentration of oxygen, and it is used for the sickest breathing patients. A BVM connected to oxygen, with a reservoir attached where one is fitted, delivers the highest concentration to a patient who cannot breathe for themselves. Your protocol and medical direction set the choice, the flow, and the target for each patient. Oxygen is treated as a medicine in the United States. Giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Open and clear the airway before you ventilate, make a firm seal, and give gentle breaths until the chest rises.

See also: Entry 2.2 (airway); Entry 3.4 (CPR); Entry 6.1 (child sizing and rate).

2.4 Respiratory Distress: Asthma, COPD and Pneumonia

Breathing difficulty is one of the most common reasons people call for help, and its causes range from a minor chest cold to a life-threatening attack. This entry gives you a way to judge how serious it is and covers three common causes.

WHAT IT IS. Respiratory distress is difficult or labored breathing. Respiratory failure is the stage at which breathing can no longer meet the body's needs, and it can follow distress quickly,

particularly in a tiring patient. Signs of distress include sweating, using the neck and shoulder muscles to breathe, and belly breathing. A rate that is high or rising is a warning, and Entry 1.4 gave typical resting values. Signs of failure include drowsiness or confusion, weak or tiring breathing effort, a very quiet chest in a patient who was wheezing, and blue lips.

WHY IT MATTERS. A patient who is working hard to breathe can tire, and a tiring patient can stop. The signs of tiring are subtle: the patient becomes calmer and quieter, which is easy to mistake for improvement. Reassessing breathing rate, effort and mental state at short intervals catches the change.

REASSESSING A BREATHING PATIENT. Because the patient's condition can change quickly, set a rhythm for rechecking. Each time, note the breathing rate, how many words the patient can say at once, whether the neck and shoulder muscles are working, the color of the lips and the level of alertness. Write each set down with the time, as Entry 1.5 describes. A patient who needs more help than before is worsening even if each single number looks acceptable. A patient who is tiring may change position, lean forward on outstretched arms and start to say fewer words.

ASTHMA. Asthma is a long-term condition in which the airways narrow, tighten and become inflamed in response to triggers such as allergens, infection, exercise or cold air. In an attack, the patient may wheeze, cough, feel tightness in the chest, and struggle to breathe. Many patients carry a reliever inhaler containing albuterol (salbutamol). A severe attack is shown by speaking only a few words at a time, a high breathing rate, visible effort, a very quiet chest, drowsiness, or no improvement after the patient's usual reliever. Sit the patient upright, keep them calm, call for

help, and help them use their own inhaler as they have been taught and as your protocol allows. Giving medicines is outside many roles; follow your scope and medical direction.

COPD. COPD is a long-term lung disease, often caused by smoking, in which the airways are narrowed and the lungs are damaged. Patients are short of breath on effort, cough, and bring up mucus, and they have flare-ups called exacerbations. Oxygen needs special care here. In some patients with COPD, a high concentration of oxygen can slow breathing and let carbon dioxide build up, so these patients need a lower oxygen target set by protocol or medical direction. Give oxygen only as your scope and medical direction allow, and watch breathing closely; do not withhold oxygen from a patient who needs it. Giving medicines is outside many roles; follow your scope and medical direction.

PNEUMONIA. Pneumonia is an infection of the lungs. It causes fever, cough, sometimes chest pain on breathing in, and a fast breathing rate. In older adults or very ill patients it may cause confusion without much cough or fever. Pneumonia can lead to sepsis, which Entry 5.4 covers. Confusion with poor breathing is a warning of its own, and Entry 5.1 returns to it. First priorities are an upright or comfortable position, oxygen as directed, early reassessment, and a call for help if breathing is hard, the patient is confused, or the lips are blue.

Figure 8 sets the same priorities out as a flow: judge severity first, and note that the severe branch calls for help before anything else.

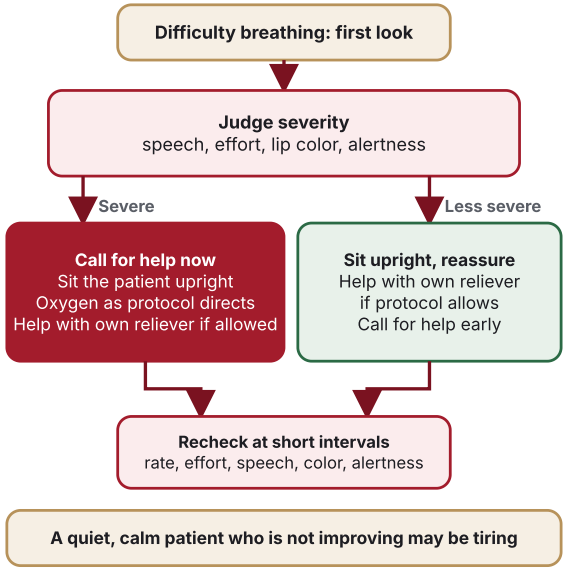


Figure 8. A first-priority flow for a patient with difficulty breathing: judge the severity, call for help, position the patient, and recheck.

RED FLAGS

Speaking only a few words at a time, a very fast breathing rate, weak or tiring effort, blue or gray lips, drowsiness or new confusion, a chest that has gone quiet, or a patient who becomes calmer without improving. Call for help at once.

SCOPE NOTE

Helping a patient use their own prescribed inhaler, and giving oxygen, depend on your role, your protocol and medical direction. If you are not sure whether you may, ask a supervisor, and always call for help in a severe attack.

NURSING AND RESPONDER PRIORITIES. Judge severity by effort, speech, color and mental state, call early for help, and reassess often because a quiet patient may be a tiring one.

See also: Entry 1.3 (primary survey); Entry 2.3 (oxygen devices); Entry 5.1 (confusion with poor breathing); Entry 5.4 (sepsis); Entry 6.1 (children).

2.5 Anaphylaxis

Anaphylaxis is a severe allergic reaction that affects the whole body and can be fatal. It tends to arrive fast, often within seconds or minutes of contact with the trigger, and it can close the airway, drop the blood pressure, or both. A bystander with no medical training may be the first person to treat it.

WHAT IT IS. In anaphylaxis, the immune system reacts to a substance it has become sensitive to, such as an insect sting, a food or a medicine, and releases chemicals that cause the airways to tighten and the blood vessels to widen. Some people have a reaction with no known cause.

WHY IT CAN KILL. Two things fail in anaphylaxis, often together. Swelling and tightening narrow the airway, so air cannot get in. Widened blood vessels leak fluid and drop the blood pressure, so the heart cannot keep the brain supplied. Either can cause collapse. Because the reaction can recur after it seems to have settled, a patient who improves still needs to be assessed by a clinician.

RECOGNITION. Symptoms can involve several body systems at once. They include hives, itching or redness of the skin, swelling of the face, eyes or tongue, difficulty breathing, wheezing, a tight chest, a hoarse or whispered voice, difficulty swallowing, abdominal pain, vomiting, dizziness or fainting, and anxiety. A warning

sign of dangerous throat swelling is a very hoarse voice or coarse noise on breathing in. When a person has a known allergy and develops breathing difficulty, throat swelling or faintness soon after exposure, treat it as anaphylaxis.

RED FLAGS

A very hoarse or whispered voice, noisy breathing in, swelling of the tongue or face, difficulty breathing, faintness or collapse, or symptoms that follow a known trigger within minutes. Call emergency services at once.

FIRST PRIORITIES. Anaphylaxis is an emergency. Call emergency services immediately. Check airway, breathing and circulation, and be ready to start CPR. Remove the trigger if you can, for example by scraping off a bee stinger. If the person has a prescribed epinephrine (adrenaline) auto-injector, help them use it at once; the first-line treatment is epinephrine, and delay is dangerous. Giving medicines is outside many roles; follow your scope and medical direction.

PROCEDURE: HELP A PATIENT USE THEIR OWN EPINEPHRINE AUTO-INJECTOR

Purpose: to help the patient receive their own prescribed, premeasured epinephrine as the device instructions direct. Scope: follows your training level, state law and local protocol.

1. Call emergency services, or ask someone to call, and stay with the patient.
2. Ask whether they carry an auto-injector, and find it; if there is none, do not delay the call, and keep checking breathing and consciousness.
3. Read the device label and follow its printed instructions exactly; do not guess or improvise.

4. Help the patient or, where your role and the law allow, use the device on the patient as the label shows.
5. Note the time the injection was given and tell the responders.
6. Place the patient in the position that suits their breathing: lying flat with legs raised if they are faint, sitting up if breathing is hard, or on the side if vomiting.
7. Keep checking breathing and consciousness, and be ready to start CPR.
8. Even if the patient improves, they need to be assessed by a clinician; symptoms can return.

> **⚠ SAFETY**
>
> Do not give anything by mouth to a person who has trouble breathing. Keep your fingers away from the needle end of the device, and put the used device in a sharps container.

STOP AND CALL FOR HELP IF breathing becomes difficult, the voice changes, swelling spreads, or the patient becomes drowsy or faint.

The steps follow the logic of the problem. Anaphylaxis is a race between a reaction that is accelerating and a treatment that works quickly, so the first moves are to bring help and the treatment together. Entry 3.5 explains how a fall in blood pressure becomes shock. Position protects the airway and the blood pressure, and the recorded time lets the next team judge the response.

NURSING AND RESPONDER PRIORITIES. Recognize the pattern early, call for help at once, help the patient use their own auto-injector as the device directs, and keep reassessing until a clinician takes over.

See also: Entry 1.3 (primary survey); Entry 2.2 (airway); Entry 3.5 (shock); Entry 6.1 (children).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Marcus was a first-year paramedic student helping at a weekend youth camp with the camp nurse, Dana. During an afternoon game, a 14-year-old camper named Leo sat down on the grass and said he could not catch his breath. Marcus reached him first, checked that the area was clear, and put on gloves.

Leo was sitting forward, resting his hands on his knees. He could say only a few words between breaths and was breathing fast. Marcus could hear a high wheeze. Leo's lips looked normal, and he was alert. Leo said he had asthma and that his inhaler was in his bag at the cabin, and that he had been running hard in the cold wind.

Marcus told Dana what he saw, and she sent a counselor for the inhaler and called emergency services at once. He kept Leo sitting upright, spoke slowly, and counted his breathing rate at the start and again 2 minutes later. It had not changed. The inhaler arrived, Dana helped Leo use it as he had been taught, and she told Marcus to keep watching his color and how much he could say. When Leo managed a full sentence, they relaxed a little. When the paramedics arrived, Dana handed over what Marcus had counted and the time of the inhaler. Leo was taken in for an assessment, because his breathing had not settled.

Care Priorities

CARE PRIORITIES

When two airway or breathing problems compete, treat the one that stops air first: an obstruction before a wheeze, and a wheeze before a low reading. Use your hands before equipment, because repositioning is quicker than fetching equipment. Treat effort as an early warning, since a patient who has stopped struggling may be tiring rather than recovering. Ask for help before the patient needs it, and say in one sentence what you see, what you have done and what you need. Decide before you start a skill what you will do if it fails, so that a failure does not surprise you at the bedside.

Review Questions

1. A patient's chest and abdomen move in opposite directions and no air can be felt at the mouth. What does this suggest?
   - **A)** Normal breathing.
   - **B)** Complete airway obstruction.
   - **C)** A mild chest cold.
   - **D)** A strong cough reflex.

2. Which technique is preferred for opening the airway of an unresponsive patient with a possible neck injury?
   - **A)** Jaw thrust.
   - **B)** Sitting the patient upright.
   - **C)** Head-tilt chin-lift.
   - **D)** Rolling the patient face down.

3 When should you use suction?

A) At all times during airway care.
B) Only after the patient wakes up.
C) Before checking for a response.
D) When you can see or hear fluid in the mouth or throat.

4 A conscious adult cannot speak, cough or breathe and points to the throat. Which action is correct?

A) Ask the person to lie down and rest.
B) Wait for the person to improve.
C) Follow your training for back blows and abdominal thrusts.
D) Begin chest compressions at once.

5 Which is a risk of squeezing a bag-valve-mask too hard or too fast?

A) Air can enter the stomach which raises the risk of vomiting.
B) The oxygen will run out.
C) The mask will fall off.
D) The patient will start breathing normally at once.

6 A patient with severe asthma becomes quieter and calmer but still has blue lips and can say only a word. What does this suggest?

A) The attack is improving.
B) The patient needs only reassurance.
C) Nothing needs to change.
D) The patient is tiring and may be failing.

7 After a bee sting, a person develops hives, a hoarse voice and difficulty breathing. What is the first priority?

A) Wash the area and watch for an hour.
B) Call emergency services and help a patient who carries one use their own epinephrine auto-injector.
C) Give a drink to restore strength.
D) Wait to see whether the symptoms pass.

8 Short case. A patient with COPD has become short of breath at home and is drowsy. List, in order, the first four things you would do and explain why oxygen needs care in this patient.